

The Standing Neutrino Splitting Ratio from First-Principles Standing Admission

Same-Scope Singleton Closure, Source-Row Numerical Readout, and One-Anchor
Mass-Spectrum Display

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Abstract

This paper gives a dependency-compressed AASC account of the standing neutrino splitting-ratio endpoint under explicit standing admission. The result is not a phenomenological fit, not a use of observed neutrino mass splittings as construction inputs for the dimensionless ratio, and not a zero-anchor derivation of absolute neutrino masses. The main result is a same-scope standing-admission numerical window for the current neutrino splitting ratio, together with a conditional one-anchor metric display of the associated projective mass-squared shape.

The argument proceeds in three typed stages. First, the standing neutrino splitting-ratio target is treated as a non-degenerate same-scope AASC regime; the UEAP branch audit eliminates every non-current route by kernel-forced claim-standing failure and leaves the current standing ratio as the unique admissible survivor. Second, an AASC standing-admission certificate is introduced for the same-scope source-row interface. From that certificate, the Lean4 formalization constructs the same-scope row admission object, the bundled numerical readout certificate, and rational decimal windows for the standing AASC coordinate, its inverse hierarchy coordinate, and the shifted hierarchy coordinate. Third, the ratio readout is continued into a projective mass-squared shape and then into a one-anchor atmospheric-gap mass-spectrum display when an external metric anchor is admitted.

The cited external source-row material is used only as provenance and row data. It is not structural proof authority. The structural authority is the AASC standing-admission certificate, the same-scope transport machinery, the singleton/readout pipeline, and the Lean4 theorem chain. Under `AASCNeutrinoSourceRowStandingAdmission`, Lean proves

$$0.0299486814926010 \leq \rho_{\nu}^{\text{current}} \leq 0.0299486814926012,$$

$$33.3904516045907 \leq (\rho_{\nu}^{\text{current}})^{-1} \leq 33.3904516045908,$$

and

$$32.8904516045907 \leq (\rho_{\nu}^{\text{current}})^{-1} - \frac{1}{2} \leq 32.8904516045908.$$

With an admitted external atmospheric-gap anchor, the same projective shape admits a one-anchor mass-spectrum display. That display is reported in the body of the paper as a conditional metric display, not as an abstract-level first-principles absolute-mass theorem. The zero-anchor absolute mass scale, source-independent internal coordinate anchor, explicit operator characteristic-polynomial syntax, PMNS numerical-coordinate derivation, and full oscillation-probability prediction remain open.

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1 Introduction: from structural Standard Model closure to neutrino readout

The AASC Standard Model spine separates structural closure from numerical parameter closure. The structural theorem concerns the recovered standalone Standard Model occupant: gauge-product organization, representation-response slots, charge lattice, anomaly survivor ledger, vacuum compatibility, Higgs-like bridge/fixation carrier, Yukawa-realizability ledger, mass-response slots, mixing/phase slots, and same-scope structural rival exhaustion. The downstream numerical program concerns values inside that occupant: mass ratios, absolute scale anchors, CKM/PMNS numerical coordinates, gauge-coupling values, Higgs-sector numerical values, and sector-specific dimensionless constants. The Standard Model structural synthesis explicitly types this separation as part of the theorem statement [1, 2].

The present paper belongs to the downstream parameter program. Its target is not the complete neutrino sector. Its target is the standing splitting-ratio object

$$\rho_\nu = \frac{\Delta m_{21}^2}{\Delta m_{31}^2}, \quad (1)$$

viewed as a same-scope AASC target over the solar and atmospheric splitting roles. The previous non-degenerate UEAP branch audit closed the admissible same-scope branch set to a singleton: the current standing ratio is the unique admissible survivor [3]. That result is an extensional closure theorem. It identifies the unique admissible occupant, but it does not, by itself, provide a lawful numerical readout or a metric mass display.

The NNR9 extension supplies the standing-admission readout interface. It introduces an AASC certificate for the same-scope source-row interface of the standing neutrino target. From that certificate, the Lean development constructs the row admission object, the bundled readout certificate, and the rational decimal windows for the standing coordinate and its inverse hierarchy presentations [4, 5, 6]. The one-anchor continuation then treats the readout as a projective mass-squared shape: one target-aligned metric anchor can display an absolute mass-squared triple while preserving the NNR9 ratio. This is not zero-anchor mass derivation. It is a lawful metric display after an explicitly typed scale anchor.

“First-principles” in the title refers to the AASC standing-admission, same-scope transport, and no-laundering constraints under which the readout is admitted. It does not mean that every displayed decimal is derived from an empty source context.

1.1 Endpoint shape

The endpoint has the following shape:

```
AASC standing source-row admission
-> same-scope NNR9 numerical readout
-> rational windows for rho, R31, and R3ell
-> one-anchor atmospheric-gap mass display when an external Delta m^2_31 anchor is
    admitted
```

The manuscript-facing Lean chain for the ratio readout is:

```
AASCNeutrinoSourceRowStandingAdmission
-> AASCNeutrinoReadoutSameScopeAdmissionRows
-> AASCNeutrinoNumericalReadoutCertificate
-> AASCNeutrinoSourceRowStandingAdmission.finalNumericalReadout
```

Mathematically, this chain says that a same-scope admitted source-row witness transports the unique standing ratio into the stated rational windows.

1.2 Main theorem spine

Theorem 1.1 (Main standing-admission and one-anchor display spine). *Assume the AASC source-row standing certificate. Then Lean constructs the same-scope admission rows and the bundled numerical readout certificate, and proves the rational windows for $\rho_\nu^{\text{current}}$, $(\rho_\nu^{\text{current}})^{-1}$, and $(\rho_\nu^{\text{current}})^{-1} - 1/2$ stated in the abstract. In addition, the AASC projective mass-squared shape supports a one-anchor reconstruction theorem: if an atmospheric mass-squared gap anchor is admitted, the reconstructed absolute mass-squared spectrum has that atmospheric gap and retains the NNR9 selected root as its gap ratio. With $\Delta m_{31}^2 = 2.513 \times 10^{-3} \text{ eV}^2$, the display values are computed in Section 13. They are not abstract-level first-principles absolute-mass claims.*

1.3 Reading rule

The words “current”, “unique”, “standing”, “readout”, and “display” are typed. “Current” means the current same-scope branch selected by the AASC branch audit. “Unique” means unique inside the admitted same-scope hybrid joint restriction, not unique across arbitrary model-building imagination. “Standing” means the object survives admissible redescription and same-scope transport. “Readout” means a lawful numerical presentation transported from admitted source rows through the AASC certificate. “Display” means a metric presentation after a scale anchor has been supplied; it does not mean the scale itself was internally derived.

2 Claim typing and nonclaims

The theorem boundary is part of the proof. The present manuscript would overclaim if read as an unconditional numerical derivation of the full neutrino sector. The exact claim is narrower: AASC first closes the same-scope standing branch, then proves conditional standing-admission numerical windows, and finally gives a one-anchor mass display when a target-aligned external atmospheric-gap anchor is admitted.

Definition 2.1 (Status labels). The paper uses four status labels.

1. **Proved/module-supported:** established by an upstream AASC support theorem or Lean object and imported as a module output.
2. **Lean-mapped:** corresponding Lean declarations are named and used as the formal endpoint for the manuscript claim.
3. **Conditional:** the result follows from a declared certificate, admission object, or metric anchor.
4. **Open:** exact obstruction is named but not closed in the present manuscript.

Status 2.2 (Closed claims). The following claims are closed at the level stated.

- C1.** The standing neutrino splitting-ratio branch audit is closed to a same-scope singleton under the non-degenerate UEAP kernel-operationalization regime.
- C2.** The row-readout endpoint is represented by an explicit AASC standing-admission certificate rather than a hidden row-admission label.

- C3.** Conditional on that certificate, Lean constructs the same-scope admission rows and derives the rational windows for ρ_ν , R_{31} , and $R_{3\ell}$.
- C4.** The coordinate convention between the AASC standing coordinate and the printed external hierarchy coordinate is explicitly typed: the printed coordinate is the inverse coordinate.
- C5.** The AASC source rows instantiate a projective mass-squared shape whose gap ratio is the NNR9 selected root.
- C6.** Conditional on an admitted atmospheric-gap metric anchor, Lean reconstructs an absolute mass-squared spectrum preserving the selected root as the gap ratio.

Nonclaim 2.3 (Nonclaims). This paper does not claim:

- N1.** a zero-anchor derivation of absolute neutrino masses;
- N2.** a zero-anchor derivation of the absolute neutrino mass scale;
- N3.** a derivation of PMNS numerical coordinates;
- N4.** a derivation of the full oscillation-probability table;
- N5.** that the external source package is proof authority;
- N6.** that observed neutrino mass splittings are used as construction inputs for the ratio readout;
- N7.** that branch-singleton closure alone supplies a number;
- N8.** that the endpoint is unconditional from empty context;
- N9.** that the external inverse coordinate and the AASC standing coordinate are the same coordinate;
- N10.** that the NuFIT atmospheric-gap anchor is derived by AASC.

Remark 2.4 (Conditionality). The theorem is unconditional only relative to the AASC source-row standing-admission certificate. An AASC-internal replacement for the source-row law would strengthen the result from admitted-source readout to source-independent internal derivation. This distinction is retained throughout the manuscript rather than hidden in implementation terminology.

3 Endpoint status box: achieved, conditional, and open

Achieved under AASC standing admission. The current standing splitting ratio has certified rational windows; the inverse hierarchy coordinate and shifted hierarchy coordinate are derived report coordinates.

Achieved as conditional one-anchor display. The admitted projective mass-squared shape gives an absolute mass-squared display after a target-aligned external atmospheric-gap anchor is supplied.

Open. Zero-anchor absolute mass scale, source-independent internal coordinate anchor, explicit operator characteristic-polynomial syntax, PMNS numerical-coordinate derivation, and full oscillation-probability prediction remain outside the present endpoint.

Object	Status	Boundary
Standing splitting ratio ρ_ν	achieved conditionally	AASC source-row standing admission required
Inverse hierarchy R_{31}	derived display	reciprocal of ρ_ν
Shifted coordinate $R_{3\ell}$	derived display	$R_{31} - 1/2$
Projective mass-squared shape	achieved conditionally	shape, not absolute scale
Absolute mass-squared spectrum	one-anchor display	requires external Δm_{31}^2 anchor
Displayed neutrino masses	one-anchor display	computational digits conditional on the chosen anchor
Zero-anchor absolute mass scale	open	no internal scale carrier supplied here
PMNS numerical coordinates	not claimed	witness scope is not a PMNS-value derivation

4 Core vocabulary and coordinate convention

4.1 The standing coordinate

The AASC standing coordinate is

$$\rho_\nu = \frac{\Delta m_{21}^2}{\Delta m_{31}^2}. \quad (2)$$

Here Δm_{21}^2 is the solar mass-squared splitting role and Δm_{31}^2 is the atmospheric mass-squared splitting role. The ratio is a dimensionless same-scope object. The theorem is about this object and its inverse hierarchy presentations.

4.2 The printed inverse coordinate

The cited source-row package prints the inverse hierarchy coordinate

$$R_{31} = \frac{\Delta m_{31}^2}{\Delta m_{21}^2}. \quad (3)$$

It also prints the shifted coordinate

$$R_{3\ell} = R_{31} - \frac{1}{2}. \quad (4)$$

Thus the printed hierarchy coordinate is not the same coordinate as the AASC standing ratio. The bridge is

$$\rho_\nu = R_{31}^{-1} = (R_{3\ell} + 1/2)^{-1}. \quad (5)$$

Lean records this convention through:

```
aascNeutrinoR31_eq_R3ell_plus_half
aascNeutrinoStandingRatioReadout_eq_inv_R31
aascNeutrinoStandingRatioReadout_eq_inv_R3ell_plus_half
```

Mathematically, these identities state that the source's printed hierarchy coordinate is the reciprocal presentation of the AASC standing coordinate, with a further shifted display coordinate.

4.3 Projective shape coordinate

A projective mass-squared shape is a mass-squared triple up to common scale where the ratio of adjacent splitting roles is preserved. In the source-row display, the levels are

$$L_1 = 1.0120134524368151586, \quad (6)$$

$$L_3 = 159.3652713329476282194, \quad (7)$$

$$L_5 = 5288.4988071409118531250. \quad (8)$$

The index labels 1, 3, 5 belong to the source-row expression; the three physical levels are the three entries of the projective triple. The AASC ratio is

$$\frac{L_3 - L_1}{L_5 - L_1} = 0.0299486814926011 \dots \quad (9)$$

5 Dependency ledger

Layer	Output used here	Status
AASC kernel	admissibility, standing, reference, non-repair discipline	proved/module-supported
UEAP branch audit	non-current branches fail kernel-forced UEAP coordinates	module-supported, Lean-mapped
Hybrid singleton	unique survivor in the hybrid joint restriction	module-supported, Lean-mapped
Source-row prove-nance	finite source rows, coefficient pipeline, coordinate convention	row data/provenance only
Standing admission	source rows admitted as same-scope standing carrier	conditional on named AASC source-row certificate
Numerical readout	rational windows for ρ_ν , R_{31} , $R_{3\ell}$	conditional Lean endpoint
Projective shape	mass-squared shape preserving selected ratio	Lean-mapped continuation
One-anchor display	metric mass-squared spectrum from admitted Δm_{31}^2 anchor	conditional metric display
Zero-anchor scale	internally fixed absolute neutrino scale	open
PMNS values	numerical PMNS coordinates	not claimed

Certificate 5.1 (Dependency discipline).

The paper uses external row data only after the AASC standing-admission layer. The row data do not replace the AASC proof obligations. Conversely, the AASC branch singleton does not by itself produce the decimal windows or the one-anchor mass display. The three roles are separate: branch closure, source-row admission/readout, and metric display.

6 Standing neutrino ratio target

Let C be the standing neutrino splitting-ratio certificate. It contains two non-collapsed response roles, the solar and atmospheric splitting roles, together with a common carrier, a fixed atmospheric

denominator, a subordinate solar-to-atmospheric relation, and the reportable standing ratio. Let M be the MR3 spectral-source admission and let H be the hybrid modular/spectral/scoto compression network. The hybrid joint restriction is the class of ratios r satisfying

$$r \in C.\text{inMinimalInterval}, \quad H_{\text{mod}}(r), \quad H_{\text{spectral}}(r), \quad H_{\text{scoto}}(r). \quad (10)$$

The current branch is **currentStandingRatio**. Non-current branches include modular-only, scoto-parameter-fiber, spectral-without-finite-collapse, sterile-scope-extension, phase-topology-breaking, matter-transport-breaking, absolute-mass-import, Dirac/Majorana-selector, and identity-selector routes [3].

Definition 6.1 (Standing splitting-ratio target). The standing neutrino splitting-ratio target is the same-scope dimensionless object

$$\rho_\nu = \frac{m_2^2 - m_1^2}{m_3^2 - m_1^2}, \quad (11)$$

where the numerator and denominator are not raw empirical numbers but the solar and atmospheric splitting roles as carried inside the admitted AASC target.

Remark 6.2 (Not an absolute mass target). The target is invariant under common mass-squared translation and common nonzero scaling where the affine quotient is defined. Therefore it cannot determine an absolute offset or absolute mass scale by itself. This is why the one-anchor display is a display theorem, not a zero-anchor mass theorem.

7 UEAP as kernel operationalized

The UEAP claim-standing audit is not an optional additional filter. In the non-degenerate regime used here, UEAP is the admissibility kernel expressed as auditable claim-standing coordinates. A branch that purports to survive the hybrid joint restriction must be UEAP sigma-legitimate for its exposed branch class. A non-current branch fails exactly when the kernel forces failure of at least one legitimacy coordinate: target fixation, carrier adequacy, fixed meaning, fixed scope, target-carrier alignment, exhaustion modulo skin, laundering exclusion, ancestry separation, boundary declaration, evidence-network closure, or report preservation.

Theorem 7.1 (Non-degenerate UEAP branch closure). *In the admitted non-degenerate same-scope neutrino regime, the hybrid joint restriction has no non-current survivor. Its unique survivor is the current standing ratio.*

Proof spine. If a ratio in the hybrid joint restriction has current route class, current-class point identification gives the current standing ratio. If it has non-current route class, survival in the hybrid joint restriction makes it a claim-standing route. Since UEAP is the kernel operationalized, the branch must be sigma-legitimate. But each non-current branch has a kernel-forced UEAP coordinate failure recorded in the branch ledger. A coordinate failure blocks sigma-legitimacy. Therefore no non-current branch survives and the current branch is the unique survivor. $\square$

The Lean-side proof path is represented by:

```
AASCSNondegenerateUEAPKernelOperationalization
-> AASCUEAPCitedCoordinateBranchLedger
-> AASCUEAPCoordinateBranchRegistry
-> AASCUEAPBranchAudit
-> AASCBBranchImpossibilityAudit
-> HybridJointRestrictionSingleton
```

Mathematically, this path says that every non-current route is excluded by a typed claim-standing failure, while the current route is point-identifying.

8 Why singleton closure is not yet numerical evaluation

Branch singleton closure identifies the admissible point extensionally. It does not by itself display the point as a rational decimal window, a closed expression, an algebraic root, or an absolute mass spectrum. The NNR9 evaluator machinery supplies the additional readout layer.

The historical evaluation frontier stated the blocker as:

```
NoNumericalEvaluatorForCurrentStandingBranch C
CurrentStandingRatioOpaqueToNumericReadout C
```

The current endpoint resolves that blocker through a standing-admission readout route. The earlier frontier remains useful as a classification device: it distinguishes normal-form-only, finite-class, algebraic-class, closed-expression, internal-coordinate exact-value frontier, and decimal-window strengths. The present paper claims standing-admission numerical windows and one-anchor metric display, not an unconditional zero-anchor mass theorem.

9 External source provenance and source-row law

The external source package used for row provenance is:

M. Pajuhaan, *A closed, fit-free prediction for the neutrino mass hierarchy from the charged-lepton emergent pipeline*, Zenodo, DOI 10.5281/zenodo.17169926.

The package contains a paper file and a companion Python source. In the local audit, the relevant source lines are the contamination audit, the $P_{1/R}$ definition, the $k = 1, 3, 5$ row construction, and the printed hierarchy readouts.

The row law translated into Lean is

$$P_{1/R}(k) = 1 + \beta k^2 + \delta k^4, \quad k \in \{1, 3, 5\}. \quad (12)$$

The finite-decimal source readback is

$$\beta = 0.0000780161359733586, \quad \delta = 0.0119354363008418, \quad s = 1. \quad (13)$$

The mass-squared row expression is

$$L(k) = k^4 P_{1/R}(k). \quad (14)$$

Lean records these as:

```
aascNeutrinoBeta
aascNeutrinoDelta
aascNeutrinoScale
aascNeutrinoP10verR
aascNeutrinoMassSquaredLevel
aascNeutrinoNumeralParameterTable
```

Mathematically, these declarations fix the row-data presentation used by the AASC standing-admission interface. They do not, by themselves, confer standing authority.

Certificate 9.1 (Provenance boundary).

The source law is not admitted because it is printed externally. It enters only through the AASC standing-admission interface. The source package supplies row data and provenance; the standing authority is the AASC certificate.

Remark 9.2 (Provenance risk). The present paper does not claim that the external source independently proves the neutrino ratio. The AASC claim is conditional: if the cited row law is admitted by the AASC source-row standing certificate, then the same-scope readout follows. An independent AASC-internal replacement for the source-row law remains a frontier target.

10 AASC standing-admission certificate

The broad row-admission object is:

AASCNeutrinoReadoutSameScopeAdmissionRows

In the submitted endpoint it is not assumed directly. It is produced from the stronger corpus-shaped certificate:

AASCNeutrinoSourceRowStandingAdmission

The certificate names the finite PMNS invariant-witness scope, mass-squared splitting witness standing, same-scope oscillation rival exhaustion, bridge-anchor admissibility, evidence-ledger completeness, same-standing source-row carrier, quotient/skin transport compatibility, non-empirical source-row provenance, QNF alignment, hybrid singleton, and strict mass-squared ordering.

10.1 Certificate-field discharge table

Certificate field	What it blocks	Corpus/source support	Lean support
Finite PMNS invariant-witness scope	PMNS numerical-value overclaim	NNR source-row admission and finite witness-scope discipline	Source-row admission
Mass-squared splitting witness standing	Raw empirical gap import	Standing splitting-role ledger and no empirical splitting import audit	Splitting witness route
Same-scope oscillation rival exhaustion	Alternative same-target route survival	UEAP branch audit and hybrid singleton closure	Hybrid singleton
Bridge-anchor admissibility	Row-data laundering	AASC bridge-anchor row-readback discipline	Bridge audit route
Evidence-ledger completeness	Incomplete readout support	NNR9 source-row standing discharge ledger	Readback audit
Same-standing carrier readback	Source rows as proof authority	Source rows transported through admitted same-standing carrier	Row admission map
Quotient/skin compatibility	Notation or representative drift	QNF alignment and quotient-skin transport controls	QNF route
No empirical mass/splitting import	Observed-value backfit	Source contamination/no-import audit	No-import route
No hidden selected-root encoding	Root smuggling	Coefficient/root encoding audit	No-hidden-root route
Level row equality	Disconnected source law	Equality of levels 0,1,2 with k=1,3,5 rows	Row-level map
QNF alignment and hybrid singleton	Disconnected evaluator	QNF/root alignment plus singleton branch closure	Final readout route

Strict mass-squared ordering
Untyped orientation

Ordering/interface assumption in- Ordering route
side source-row certificate

The source-row carrier field is the critical interface: it asserts not merely numerical agreement but same-standing transport of the row law into the neutrino splitting-ratio target. Without this field, the row data would remain external provenance and would not support the readout theorem.

The discharge theorem is:

```
AASCNeutrinoSourceRowStandingAdmission.discharge_sameScopeAdmissionRows
```

The correct statement is therefore:

```
Given AASCNeutrinoSourceRowStandingAdmission, Lean constructs
AASCNeutrinoReadoutSameScopeAdmissionRows and then proves the final numerical windows.
```

Without `AASCNeutrinoSourceRowStandingAdmission`, the Pajuhaan/Zenodo rows remain external provenance and no AASC numerical readout theorem follows. This is the central conditional premise of the numerical readout layer, not a hidden unconditional assumption.

11 Lean4 formalization status and reproducibility

The formalization route is located in:

```
MaleyLean.Papers.Neutrino.SourceTheorems.MathlibSpectralOperator
```

The reported build command is:

```
lake build MaleyLean.Papers.Neutrino
```

The formalization audit reports that this build passes, that no `sorry`, `admit`, `axiom`, or `unsafe` token occurs in the relevant Lean file, and that the only remaining output is a pre-existing unrelated `simp` linter warning [4, 5]. This is a formalization-status statement about the Lean file and build route. It is not an assertion that the external source is itself proof authority.

11.1 Reproducibility ledger

Item	Value
Repository	somamaley-ux/AASC-REPOS
Repository URL	https://github.com/somamaley-ux/AASC-REPOS
Repository commit SHA	3d538abd3bedf20fc8973f20783da52d31acaccf
Lean toolchain	leanprover/lean4:v4.28.0
Lake project name	BivalenceNonDegenerateReasoning
Default Lake target	MaleyLean
mathlib source	https://github.com/leanprover-community/mathlib4.git
mathlib revision	8f9d9cff6bd728b17a24e163c9402775d9e6a365
Build command	lake build MaleyLean.Papers.Neutrino
Primary theorem file	MaleyLean.Papers.Neutrino.SourceTheorems.MathlibSpectralOperator

Primary theorem file SHA256	4E7DD41072CC1EC6E8C3E34F082FE84E79A6E777B3D970 5389E1329F863B77F3
Scan command	listed below
Archive DOI / release record	10.5281/zenodo.20441858

The manuscript reports no matches in the primary theorem file at the recorded SHA256 for the scan command below. This proof-placeholder scan is reported for the primary theorem file named in the reproducibility ledger; dependency-wide axiom auditing is a separate artifact-level check unless explicitly stated.

```
grep -nE '\b(sorry|admit|axiom|unsafe)\b' \
  MaleyLean/Papers/Neutrino/SourceTheorems/MathlibSpectralOperator.lean
```

Remark 11.1 (Repository scope). The repository metadata identifies AASC-REPOS as the Lean4 formalization repository for the AASC paper family. The reproducibility ledger records the commit and primary theorem-file hash for the submitted endpoint. The Zenodo DOI is the release/archive anchor for the cited formalization package: *AASC formalization archive*, version associated with the recorded repository commit and primary theorem-file hash.

12 Numerical readout theorem

The source-row computation gives the certified rational windows

$$0.0299486814926010 \leq \rho_\nu \leq 0.0299486814926012, \quad (15)$$

$$33.3904516045907 \leq R_{31} \leq 33.3904516045908, \quad (16)$$

$$32.8904516045907 \leq R_{3\ell} \leq 32.8904516045908. \quad (17)$$

Lean exposes these by:

```
aascNeutrinoSourceGapRatio_in_decimalWindow
aascNeutrinoSourceInverseHierarchyRatio_in_decimalWindow
aascNeutrinoSourceR3ellRatio_in_decimalWindow
```

After same-scope AASC admission, the selected root inherits the corresponding windows through:

```
aascNeutrinoNumericalReadoutCertificate_selectedRoot_in_decimalWindow
aascNeutrinoNumericalReadoutCertificate_selectedRoot_inverse_in_decimalWindow
aascNeutrinoNumericalReadoutCertificate_selectedRoot_r3ell_in_decimalWindow
```

Mathematically, the selected same-scope root is transported into a certified rational window for the standing ratio, and the inverse and shifted windows are then report-coordinate presentations.

Theorem 12.1 (Standing-admission numerical readout). *Given the AASC source-row standing certificate, Lean proves:*

$$0.0299486814926010 \leq \rho_\nu^{\text{current}} \leq 0.0299486814926012, \quad (18)$$

$$33.3904516045907 \leq (\rho_\nu^{\text{current}})^{-1} \leq 33.3904516045908, \quad (19)$$

$$32.8904516045907 \leq (\rho_\nu^{\text{current}})^{-1} - \frac{1}{2} \leq 32.8904516045908. \quad (20)$$

The theorem name is:

```
AASCNeutrinoSourceRowStandingAdmission.finalNumericalReadout
```

Remark 12.2 (Coordinate discipline). The first window is the AASC standing coordinate. The second and third windows are report coordinates derived from inversion and shift. Confusing them changes the claim.

13 One-anchor continuation and absolute mass-squared display

The endpoint includes a lawful one-anchor continuation. This section states it separately to prevent a zero-anchor misreading.

13.1 Ratio-only residual family

Lean records the ratio-only residual family

$$m_1^2 = t, \quad m_2^2 = t + \rho D, \quad m_3^2 = t + D. \quad (21)$$

For $D \neq 0$, this family has splitting ratio ρ . Changing the offset t does not change the ratio. The relevant Lean objects are:

```
threeFlavorRatioOffsetGapFamily_gapRatio
threeFlavorRatioOffsetGapFamily_offsetInvariant
```

Proposition 13.1 (Offset guardrail). *The NNR9 ratio readout does not determine the absolute offset of the mass-squared spectrum. Therefore the ratio readout cannot be reinterpreted as a zero-anchor absolute-mass theorem.*

13.2 Projective mass-squared shape

The source-row levels define an AASC projective mass-squared shape. Lean records:

```
ThreeFlavorProjectiveMassSquaredShape
ThreeFlavorOneAnchorMassSquaredDisplayAnchor
aascNeutrinoSourceProjectiveShapeLevel
aascNeutrinoSourceProjectiveMassSquaredShape
aascNeutrinoSourceProjectiveMassSquaredShape_gapRatio_eq_source
```

This layer says that the AASC-admitted source rows instantiate a projective mass-squared shape and that its gap ratio is the NNR9 source ratio. It still does not fix an absolute metric scale.

13.3 One-anchor metric input

The MR6-style positive continuation is represented by:

```
AASCNeutrinoOneAnchorMetricInput
AASCNeutrinoOneAnchorMetricInput.toDisplayAnchor
aascNeutrinoOneAnchorSpectrumReconstruction
aascNeutrinoNNR100OneAnchorMassSpectrumDisplay
aascNeutrinoOneAnchorSpectrum_gapRatio_eq_NNR9GapRatio
aascNeutrinoOneAnchorSpectrum_gapRatio_eq_selectedRoot
aascNeutrinoOneAnchorSpectrum_anchorLevel_eq_anchorValue
AASCNeutrinoOneAnchorAbsoluteMassSquaredSpectrumAvailable
aascNeutrinoOneAnchorMetricInput_discharge_absoluteSpectrumNeed
```

Mathematically, a projective mass-squared shape plus one target-aligned metric anchor reconstructs an absolute mass-squared triple, preserves the selected gap ratio, and fixes the anchored component.

Theorem 13.2 (One-anchor reconstruction). *Given the AASC projective mass-squared shape and one target-aligned metric display anchor, Lean constructs a mass-squared triple (m_1^2, m_2^2, m_3^2) such that the anchored component equals the admitted anchor value and the reconstructed gap ratio equals the NNR9 selected root.*

13.4 Atmospheric-gap specialization

The preferred neutrino scale anchor is the atmospheric mass-squared gap. Lean records:

```
AASCNeutrinoAtmosphericGapMetricAnchor
aascNeutrinoAtmosphericGapScale
aascNeutrinoAtmosphericGapAnchoredMassSquared
aascNeutrinoAtmosphericGapAnchored_atmosphericGap_eq_anchor
aascNeutrinoAtmosphericGapAnchored_gapRatio_eq_selectedRoot
AASCNeutrinoAtmosphericGapAnchoredSpectrumAvailable
aascNeutrinoAtmosphericGapMetricAnchor_discharge_spectrumNeed
```

The scale is

$$\lambda = \frac{\Delta m_{31}^2}{L_5 - L_1}, \quad (22)$$

and the displayed mass-squared spectrum is

$$m_i^2 = \lambda L_i. \quad (23)$$

Lean proves that the displayed atmospheric gap equals the admitted anchor and that the displayed gap ratio remains the NNR9 selected root.

13.5 Worked atmospheric-gap display

For a concrete metric display, take the normal-ordering atmospheric-gap central value used here as a display anchor

$$\Delta m_{31}^2 = 2.513 \times 10^{-3} \text{ eV}^2. \quad (24)$$

This is an external display anchor, cited from the NuFIT three-flavor oscillation fit central value used here as a display anchor [11]. It is not derived internally by AASC. The worked display uses the normal-ordering atmospheric-gap convention. It is not a proof that inverted ordering is impossible unless a separate AASC ordering/interface theorem is invoked. The notation m_1, m_2, m_3 in the display denotes the ordered three-level display induced by the admitted source-row convention; it is not a separate empirical mass-ordering fit.

Using

$$L_1 = 1.0120134524368151586, \quad (25)$$

$$L_3 = 159.3652713329476282194, \quad (26)$$

$$L_5 = 5288.4988071409118531250, \quad (27)$$

we obtain

$$\lambda = \frac{2.513 \times 10^{-3}}{5288.4988071409118531250 - 1.0120134524368151586} = 4.75273054676883 \times 10^{-7}. \quad (28)$$

Then

$$(m_1^2, m_2^2, m_3^2) = \lambda(L_1, L_3, L_5). \quad (29)$$

The displayed masses are

$$m_1 = 0.000693529181010968 \text{ eV} = 0.693529181010968 \text{ meV}, \quad (30)$$

$$m_2 = 0.00870298910236135 \text{ eV} = 8.70298910236135 \text{ meV}, \quad (31)$$

$$m_3 = 0.0501346285787071 \text{ eV} = 50.1346285787071 \text{ meV}. \quad (32)$$

The sum is

$$\sum_i m_i = 0.0595311468620794 \text{ eV} = 59.5311468620794 \text{ meV}. \quad (33)$$

The display checks are

$$\Delta m_{31}^2 = 0.002513 \text{ eV}^2, \quad (34)$$

$$\Delta m_{21}^2 = 7.52610365909066 \times 10^{-5} \text{ eV}^2, \quad (35)$$

$$\frac{\Delta m_{21}^2}{\Delta m_{31}^2} = 0.0299486814926011. \quad (36)$$

13.6 Anchor-uncertainty propagation

The displayed absolute masses inherit the uncertainty of the admitted atmospheric-gap anchor. For this one-anchor display,

$$m_i \propto \sqrt{\Delta m_{31}^2}, \quad \frac{\delta m_i}{m_i} \simeq \frac{1}{2} \frac{\delta \Delta m_{31}^2}{\Delta m_{31}^2}. \quad (37)$$

Using the NuFIT normal-ordering anchor value in Eq. (24), with the reported one-sigma interval

$$\Delta m_{31}^2 = (2.513_{-0.019}^{+0.021}) \times 10^{-3} \text{ eV}^2 \quad (38)$$

and the three-sigma range

$$2.451 \times 10^{-3} \leq \Delta m_{31}^2 \leq 2.578 \times 10^{-3} \text{ eV}^2, \quad (39)$$

the anchor-only propagated display intervals are:

Quantity	Central value	One-sigma anchor-propagated interval	Three-sigma anchor-propagated interval
m_1	0.000693529 eV	0.000690902–0.000696421 eV	0.000684920–0.000702441 eV
m_2	0.008702989 eV	0.008670026–0.008739277 eV	0.008594960–0.008814824 eV
m_3	0.050134629 eV	0.049944743–0.050343669 eV	0.049512313–0.050778868 eV
$\sum_i m_i$	0.059531147 eV	0.059305672–0.059779367 eV	0.058792193–0.060296133 eV

Equivalently,

$$\sum_i m_i = 0.05953_{-0.00023}^{+0.00025} \text{ eV} \quad (1\sigma, \text{ anchor only}), \quad (40)$$

$$\sum_i m_i = 0.05953_{-0.00074}^{+0.00076} \text{ eV} \quad (3\sigma, \text{ anchor only}). \quad (41)$$

The corresponding anchor-only solar-splitting display is

$$\Delta m_{21}^2 = 7.52610 \times 10^{-5} \text{ eV}^2, \quad (42)$$

$$\Delta m_{21}^2 \in [7.46920, 7.58899] \times 10^{-5} \text{ eV}^2 \quad (1\sigma, \text{ anchor only}), \quad (43)$$

$$\Delta m_{21}^2 \in [7.34042, 7.72077] \times 10^{-5} \text{ eV}^2 \quad (3\sigma, \text{ anchor only}). \quad (44)$$

These intervals propagate only the admitted atmospheric-gap anchor uncertainty. The slight asymmetry of the one-sigma propagated mass intervals reflects the asymmetric uncertainty quoted for the admitted atmospheric-gap anchor. They are not zero-anchor AASC mass uncertainties, and they do not include an independent uncertainty budget for the AASC standing-admission ratio window. The ratio window is much narrower than the displayed anchor uncertainty and is not the limiting term for this one-anchor mass-spectrum display.

Remark 13.3 (Display precision and uncertainty). The displayed digits are computational display digits conditional on the chosen atmospheric-gap anchor. They are not a claim that the physical neutrino masses are known to all displayed digits. The uncertainty intervals above are anchor-propagated display intervals: they inherit the external NuFIT atmospheric-gap uncertainty and leave the dimensionless AASC standing ratio unchanged.

Certificate 13.4 (One-anchor display certificate).

The displayed mass values are certified as one-anchor metric display values. The AASC contribution is the standing ratio, projective shape, source-row admission, and Lean-checked transport. The atmospheric gap supplies an external metric scale. If a different external atmospheric-gap convention is selected, the displayed masses rescale by the square root of the anchor ratio, while the dimensionless AASC ratio remains unchanged.

14 Metric display is not structural proof authority

A possible misreading is that the paper imports the NuFIT atmospheric gap and derives nothing. That is not the proof structure. The NuFIT value enters only after the dimensionless AASC standing ratio has already been read out and after the projective source-row shape has been admitted. It functions as a scale anchor for display.

The proof layering is:

```
AASC standing source-row admission -> rho window
AASC source-row projective shape -> dimensionless mass-squared shape
external atmospheric-gap anchor -> metric mass display
```

The third step does not authorize the first two. It only supplies a unit scale.

Proposition 14.1 (Scale-anchor separation). *Let Δ be the admitted atmospheric-gap anchor and let $\lambda = \Delta/(L_5 - L_1)$. Replacing Δ by Δ' rescales all displayed masses by $\sqrt{\Delta'/\Delta}$, but leaves $(L_3 - L_1)/(L_5 - L_1)$ unchanged. Therefore the AASC dimensionless readout is independent of the external metric anchor.*

15 Evaluation frontier: achieved and still open

The evaluation frontier has changed. The old blocker was that branch singleton closure did not evaluate the current branch. The current readout pipeline has inhabited the standing-admission evaluator and the one-anchor display route. The remaining frontier concerns stronger forms of internal anchoring and zero-anchor scale closure.

Object	Current state	Comment
Branch singleton	achieved	current branch uniquely survives

Standing ratio windows	achieved conditionally	from named AASC source-row certificate
Inverse hierarchy display	achieved conditionally	reciprocal and shifted coordinates
Projective mass-squared shape	achieved conditionally	from admitted source rows
One-anchor atmospheric mass display	achieved conditionally	requires external Δm_{31}^2 anchor
Closed symbolic gap quotient	active	$(m_2^2 - m_1^2)/(m_3^2 - m_1^2)$
Internal coordinate anchor independent of row admission	open	non-tautological coordinate-anchor theorem
Explicit operator characteristic-polynomial syntax	open	stronger MR3 route
Zero-anchor scale	open	no scale carrier supplied here
PMNS values	not claimed	outside this endpoint

The strongest closed endpoint is:

```
standing-admission numerical splitting-ratio readout
+ projective mass-squared shape
+ one-anchor atmospheric-gap mass-spectrum display
```

The strongest still-open endpoint is:

```
zero-anchor absolute neutrino mass-scale derivation
```

16 MR3 characteristic readback and algebraic-class frontier

MR3 supplies a frontier route, not the present endpoint. MR3 is not used to obtain the standing-admission decimal windows reported in the main theorem; it records a stronger future algebraic/operator-frontier route. The current MR3 route supports a carrier-level shape-class evaluator: minimal shape-class readback is available, while explicit operator characteristic-polynomial syntax remains open [7]. This section records the status only; detailed object names are moved to Appendix C.

Status 16.1 (MR3 current status). The minimal MR3 route supplies a carrier-level shape-class evaluator. It is not yet a genuine matrix/operator characteristic polynomial, and it does not select a root or compute a numerical value.

Certificate 16.2 (MR3 frontier certificate).

The MR3 route is a future-strengthening frontier. It should not replace the main ratio-window theorem or the one-anchor display theorem. It can be reported as: minimal MR3 shape-class readback is available; explicit operator characteristic-polynomial syntax remains open.

17 Coordinate-anchor frontier

The coordinate-anchor audit remains important because it prevents a fake internal-coordinate exact-value upgrade. The current symbolic quotient is lawful, but it is not the same as a non-tautological internal coordinate anchor.

The true internal coordinate-anchor target is:

```
ThreeFlavorInternalCoordinateAnchor R A
```

It must provide an internally derived polynomial/root object and prove:

```
root = threeFlavorGapRatio A.massSquaredLevel0f
```

without defining the root by the gap quotient itself. The audit reports that no already-admitted corpus row found so far supplies this object. The corpus supplies guardrails against fake anchors: no empirical back-import, no fitted matrix texture, no raw coordinate selector, quotient/exactness discipline, role-occupancy closure, unique interior closure, parameter-normal-form discipline, and skin filtering [4].

17.1 Best source families

The current candidate families are:

- F1.** quotient normal-form coordinate law;
- F2.** operator spectral invariant;
- F3.** role-occupancy endpoint theorem strengthened from object uniqueness to coordinate uniqueness;
- F4.** modular fixed-point law translated into AASC same-scope form;
- F5.** flavor mass-sum rule made source-internal and calibration-free.

The quotient normal-form route is the best AASC-native target because it fits the existing exactness and quotient-skin machinery.

17.2 Half-side and convexity route

The audit records a useful proper-subinterval route. To prove $\rho_\nu \leq 1/2$, it is enough to prove the midpoint compression

$$2m_2^2 \leq m_1^2 + m_3^2, \quad (45)$$

or equivalently nonnegative second finite difference

$$0 \leq m_1^2 - 2m_2^2 + m_3^2. \quad (46)$$

Lean has a chain from a source curvature witness to the aligned $[0, 1/2]$ QNF interval:

```
source curvature witness
-> mass-squared convexity
-> two-step gap half dominance
-> adjacent gap half side
-> midpoint compression
-> normalized middle half compression
-> QNF lower-half criterion
-> aligned [0,1/2] numeral interval
```

This is not the main numerical readout; it is a possible internally derived inequality/frontier.

17.3 Relator even-tower socket

The audit introduces a guarded Relator even-tower socket:

$$L(k) = sk^4(1 + \beta k^2 + \delta k^4), \quad k \in \{1, 3, 5\}. \quad (47)$$

With $s \geq 0$, $\beta \geq 0$, and $\delta \geq 0$, Lean proves nonnegative second finite difference because

$$L(1) - 2L(3) + L(5) = s(464 + 14168\beta + 377504\delta). \quad (48)$$

This route is explicitly guarded. It does not accept an external Relator paper as proof authority. It creates an AASC translation socket: if the source package certifies the three-point law, same-scope targeting, fixed exponent schedule, and no empirical tuning, Lean discharges the half-side QNF interval need.

18 Theorem map

Manuscript claim	Lean object or route
External source provenance recorded	AASCNeutrinoCoefficientPipelineSourceDerived
No empirical coefficient import	AASCNeutrinoNoMassOrSplittingInputInCoefficientPipeline
No hidden selected-root encoding	AASCNeutrinoNoHiddenSelectedRootEncodingInSourceRows
Concrete coefficient table	aascNeutrinoNumeralParameterTable
AASC row law	aascNeutrinoP10verR, aascNeutrinoMassSquaredLevel
Coordinate inversion	aascNeutrinoStandingRatioReadout_eq_inv_R31
Source ρ window	aascNeutrinoSourceGapRatio_in_decimalWindow
Source R_{31} window	aascNeutrinoSourceInverseHierarchyRatio_in_decimalWindow
Source $R_{3\ell}$ window	aascNeutrinoSourceR3ellRatio_in_decimalWindow
AASC standing admission certificate	AASCNeutrinoSourceRowStandingAdmission
Discharge of broad row socket	AASCNeutrinoSourceRowStandingAdmission.d ischarge_sameScopeAdmissionRows
AASC row admission produced	AASCNeutrinoReadoutSameScopeAdmissionRows
Bundled readout certificate	AASCNeutrinoNumericalReadoutCertificate
Final numerical theorem	AASCNeutrinoSourceRowStandingAdmission.f inalNumericalReadout
Projective shape	aascNeutrinoSourceProjectiveMassSquaredShape
One-anchor reconstruction	aascNeutrinoOneAnchorSpectrumReconstruction

Atmospheric-gap anchored display	<code>aascNeutrinoAtmosphericGapAnchoredMassSquared</code>
Anchor gap check	<code>aascNeutrinoAtmosphericGapAnchored_atmosphericGap_eq_anchor</code>
Ratio preservation under anchor	<code>aascNeutrinoAtmosphericGapAnchored_gapRatio_eq_selectedRoot</code>

19 Module certificates

Certificate 19.1 (Kernel and UEAP module).

Input: non-degenerate same-scope standing target. **Output:** UEAP as the operational form of the admissibility kernel for branch audit. **Status:** module-supported.

Certificate 19.2 (Branch audit module).

Input: hybrid modular/spectral/scoto restriction and branch ledger. **Output:** current branch singleton. **Status:** module-supported and Lean-mapped.

Certificate 19.3 (Source-row admission module).

Input: source rows, finite PMNS invariant-witness scope, mass-squared splitting witness standing, same-scope rival exhaustion, bridge-anchor admissibility, evidence-ledger completeness, quotient/skin transport compatibility, no-import guards. **Output:** `AASCNeutrinoReadoutSameScopeAdmissionRows`. **Status:** conditional on `AASCNeutrinoSourceRowStandingAdmission`.

Certificate 19.4 (Numerical readout module).

Input: same-scope admission rows. **Output:** windows for ρ_ν , R_{31} , and $R_{3\ell}$. **Status:** conditional Lean endpoint.

Certificate 19.5 (Projective shape module).

Input: admitted source levels L_1, L_3, L_5 . **Output:** projective mass-squared shape whose gap ratio equals the selected source ratio. **Status:** Lean-mapped continuation.

Certificate 19.6 (One-anchor display module).

Input: projective mass-squared shape and target-aligned atmospheric-gap metric anchor. **Output:** absolute mass-squared display preserving the selected root and matching the anchor. **Status:** conditional metric display; not zero-anchor mass derivation.

Certificate 19.7 (MR3 frontier module).

Input: MR3 source certification and native operator guardrails. **Output:** minimal shape-class evaluator; stronger operator characteristic-polynomial syntax remains open. **Status:** frontier, not main endpoint.

20 Anticipated objections and responses

Objection	Response
“You are deriving absolute neutrino masses.”	No. The paper reads out the dimensionless standing splitting ratio and gives a one-anchor metric display when an external atmospheric-gap anchor is admitted. Zero-anchor absolute mass remains open.

“The NuFIT value is imported, so the whole result is empirical.”	No. The NuFIT atmospheric gap enters only after the dimensionless ratio and projective shape are admitted. It supplies metric scale, not standing-ratio proof authority.
“A different atmospheric-gap convention changes the masses.”	Correct. The masses rescale by the square root of the anchor ratio. The dimensionless AASC ratio remains unchanged.
“Singleton closure does not give a number.”	Correct. The number comes from the standing-admission readout certificate, not from singleton closure alone.
“The source paper is being treated as proof authority.”	No. The source supplies row data and provenance. AASC standing admission supplies the structural authority.
“The result is conditional.”	Correct. It is conditional on a named AASC certificate with auditable fields.
“The paper proves PMNS values.”	No. PMNS invariant-witness scope appears inside the admission certificate; numerical PMNS coordinate derivation is not claimed.
“MR3 shape-class readback is not a characteristic polynomial.”	Correct. The minimal MR3 route gives a shape-class evaluator. Explicit operator characteristic-polynomial syntax is named as a stronger open refinement.
“The one-anchor display hides absolute scale freedom.”	No. The ratio-only residual family and offset-invariance results are explicitly stated to prevent that misreading.
“The ratio could be obtained by empirical back-import.”	The certificate includes no empirical neutrino mass/splitting import and no hidden selected-root encoding gates.

21 Open problem ledger

Open object	Meaning	Next theorem target
Zero-anchor neutrino scale	absolute mass scale without external anchor	source-derived scale carrier
Internal coordinate anchor	non-tautological polynomial/root for normalized middle coordinate	internal coordinate-anchor theorem
Explicit operator characteristic polynomial	genuine operator carrier and characteristic map	explicit operator-characteristic syntax
PMNS numerical coordinates	mixing matrix coordinate values	separate PMNS forcing module
Full oscillation probabilities	probability-level dynamical readout	separate oscillation-probability module
Narrower source-internal intervals	proof of proper subintervals like $[0, 1/2]$ from internal law	convexity/curvature/QNF half-side route

Independent placement	source-row re-	AASC-internal independent of external provenance	law row	QNF/operator/role-occupancy source theorem
-----------------------	----------------	--	---------	--

22 Conclusion

The completed endpoint is no longer merely branch selection. AASC/Lean now has a standing-admission numerical readout of the dimensionless neutrino splitting ratio, plus a projective mass-squared shape and a lawful one-anchor atmospheric-gap mass-spectrum display. The endpoint status is: achieved ratio readout, achieved conditional one-anchor display, and open zero-anchor/internal-coordinate anchor. The final ratio endpoint is:

$$0.0299486814926010 \leq \rho_{\nu}^{\text{current}} \leq 0.0299486814926012, \quad (49)$$

$$33.3904516045907 \leq (\rho_{\nu}^{\text{current}})^{-1} \leq 33.3904516045908, \quad (50)$$

$$32.8904516045907 \leq (\rho_{\nu}^{\text{current}})^{-1} - \frac{1}{2} \leq 32.8904516045908. \quad (51)$$

With the admitted atmospheric-gap anchor used in the worked display, the projective shape yields the mass-spectrum values and anchor-only propagated uncertainty intervals reported in Section 13. Those values are display values conditional on the external anchor, not the primary AASC endpoint. A source-independent internal coordinate anchor would strengthen the theorem by replacing admitted source-row readout with an internally generated coordinate law; that stronger result is not claimed here. The zero-anchor absolute mass scale remains a separate open problem.

A Appendix A: compact Lean endpoint map

```
AASCNeutrinoSourceRowStandingAdmission
AASCNeutrinoSourceRowStandingAdmission.sameScopeReadbackAudit
AASCNeutrinoSourceRowStandingAdmission.sameScopeReadbackAudit_proof
AASCNeutrinoSourceRowStandingAdmission.toSameScopeAdmissionRows
AASCNeutrinoSourceRowStandingAdmission.discharge_sameScopeAdmissionRows
AASCNeutrinoSourceRowStandingAdmission.finalNumericalReadout
AASCNeutrinoReadoutSameScopeAdmissionRows
AASCNeutrinoNumericalReadoutCertificate
```

B Appendix B: one-anchor continuation map

```
threeFlavorRatioOffsetGapFamily_gapRatio
threeFlavorRatioOffsetGapFamily_offsetInvariant
ThreeFlavorProjectiveMassSquaredShape
ThreeFlavorOneAnchorMassSquaredDisplayAnchor
aascNeutrinoSourceProjectiveMassSquaredShape
aascNeutrinoSourceProjectiveMassSquaredShape_gapRatio_eq_source
AASCNeutrinoOneAnchorMetricInput
aascNeutrinoOneAnchorSpectrumReconstruction
aascNeutrinoOneAnchorSpectrum_gapRatio_eq_selectedRoot
aascNeutrinoOneAnchorSpectrum_anchorLevel_eq_anchorValue
```

```

AASCNeutrinoAtmosphericGapMetricAnchor
aascNeutrinoAtmosphericGapAnchoredMassSquared
aascNeutrinoAtmosphericGapAnchored_atmosphericGap_eq_anchor
aascNeutrinoAtmosphericGapAnchored_gapRatio_eq_selectedRoot
AASCNeutrinoAtmosphericGapAnchoredSpectrumAvailable

```

C Appendix C: MR3 frontier object map

```

MR3CharacteristicPolynomialReadoutWitness M
MR3ShapeOutputIsCharacteristicPolynomialInvariant M
mr3CharacteristicWitnessGivesAlgebraicClassStrengthEvaluator
mr3CharacteristicWitnessGivesAlgebraicClassFrontier
mr3CorpusTheoremsGiveAlgebraicClassFrontier
ExplicitOperatorCharacteristicPolynomialSyntax
NeutrinoStandingOperatorCharacteristicPresentation
NeutrinoMassSpectralOperatorCandidate
ThreeFlavorMassSquaredSpectralOperatorData
ThreeFlavorMassSquaredSpectralSourceTheorem
ThreeFlavorMassSquaredSpectralSourceProofBundle

```

D Appendix D: recommended audit order

A hostile audit should proceed in this order:

1. Verify reproducibility metadata: repository commit, Lean toolchain, mathlib revision, build command, checked file, primary file hash, and proof-placeholder scan scope.
2. Verify that the branch audit closes the same-scope hybrid joint restriction to the current branch.
3. Verify that the source-row material is typed as row data/provenance only.
4. Verify the fields of `AASCNeutrinoSourceRowStandingAdmission`.
5. Verify the coordinate convention $\rho_\nu = R_{31}^{-1}$.
6. Verify the numerical windows and their Lean theorem names.
7. Verify that the projective mass-squared shape preserves the NNR9 selected ratio.
8. Verify that the atmospheric gap enters only as a metric display anchor.
9. Verify that zero-anchor absolute mass is not claimed.
10. Verify that MR3 characteristic readback is presented as a frontier, not as the main endpoint.

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